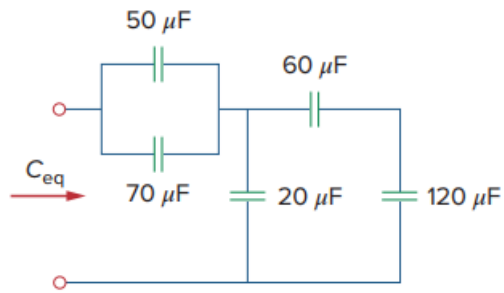


Problem

Find the equivalent capacitance seen at the terminals of the circuit in Practice Fig. 6.17.

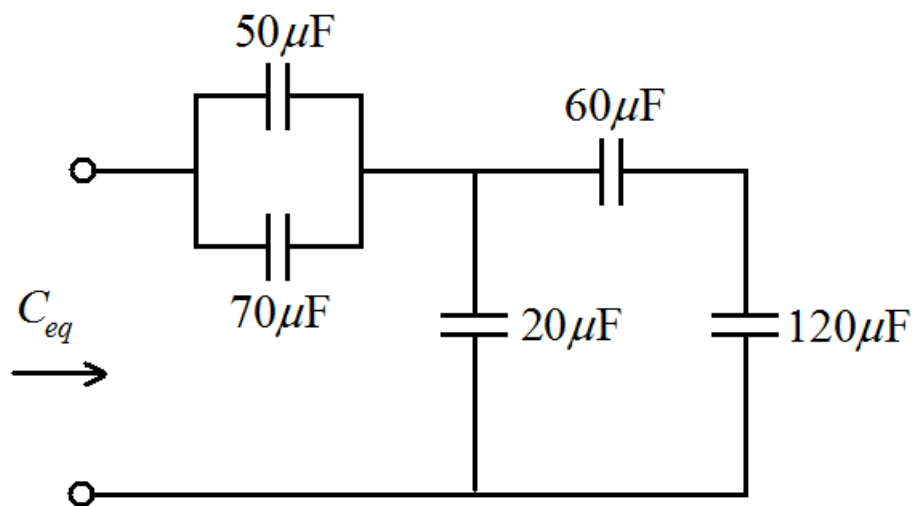
Fig. 6.17:



Step-by-step solution

Step 1 of 2

Given circuit diagram



Step 2 of 2

$120\mu\text{F}$ and $60\mu\text{F}$ capacitors are in series, their equivalent capacitance is

$$\frac{(120 \times 10^{-6}) \times (60 \times 10^{-6})}{(120 \times 10^{-6}) + (60 \times 10^{-6})} = 40\mu\text{F}$$

This $40\mu\text{F}$ capacitor is in parallel, with $20\mu\text{F}$ capacitor. Resultant combined capacitance is,

$$(40 \times 10^{-6}) + (20 \times 10^{-6}) = 60\mu\text{F}$$

This $60\mu\text{F}$ capacitor is in series with parallel combination of $50\mu\text{F}$ and $70\mu\text{F}$ capacitors.

Hence total equivalent capacitance for the entire circuit is,

$$C_{eq} = \frac{(60 \times 10^{-6})(50 \times 10^{-6} + 70 \times 10^{-6})}{(60 \times 10^{-6}) + (50 \times 10^{-6} + 70 \times 10^{-6})}$$

$$C_{eq} = \frac{(60 \times 10^{-6}) \times (120 \times 10^{-6})}{(180 \times 10^{-6})}$$

$$\therefore \boxed{C_{eq} = 40\mu\text{F}}$$